Will Thompson

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Education

University of Vermont — Burlington, VT
PhD Complex Systems and Data Science — *2022 – Present*
- GPA: 4.0/4.0

St. John's College — Santa Fe, NM
Bachelor of Arts in Liberal Arts — *2016 – 2020*
- GPA: 3.92/4.0

Experience

Johns Hopkins University — Summer 2026
Data Science for Social Good Summer Fellow — *Baltimore, MD*
- Worked with the City of Baton Rouge on early warning models for blight and structural deterioration risk for city revitalization.
- Selected from more than 400 applicants.

University of Vermont — August 2022 – Present
Graduate Research Assistant — *Burlington, VT*
- Developed a Fourier-space representation for Bayesian Neural Networks (BNNs) to enable exact, sampling-free inference and learning, bypassing computational bottlenecks of MCMC.
- Designed non-parametric Bayesian models for contagion processes, enabling the simultaneous joint reconstruction of network topology and spreading dynamics from time-series data.
- Engineered a Graph Machine Learning (GML) pipeline to infer latent interaction kernels governing stochastic network dynamics.

Los Alamos National Laboratory — June 2025 – October 2025
Graduate Research Intern — *Los Alamos, NM*
- Worked with Andrey Lokhov on inference in graphical models.
- Developed `BPTuning`, a high-performance **Julia** framework for graphical model inference on loopy graphs.
- Derived a proof of optimality for a method that optimizes effective parameters to cancel out Belief Propagation (BP) approximation errors on cyclic topologies.
- Implemented a reconstruction-based objective function to serve as a tractable proxy for exact likelihood maximization.

Zoodiker — April 2023 – January 2025
Lead Data Scientist — *Remote*
- Engineered an end-to-end NLP pipeline to process screenplay text, utilizing sentiment analysis and structural extraction algorithms to quantify narrative arcs.
- Designed anomaly detection heuristics to correlate specific character dynamics with reader engagement deviations.
- Implemented a full-stack analytics dashboard (SQL/React/Python) to visualize high-dimensional text features.
- **Patent Pending:** Filed for novel techniques in automated narrative structural analysis.

Los Alamos National Laboratory — January 2021 – August 2022
Post-Baccalaureate Researcher — *Los Alamos, NM*
- Developed a 200+ unit photon detection system for the Coherent CAPTAIN-Mills (CCM) particle physics experiment.
- Enhanced the data acquisition system (DAQ), doubling capacity and improving stability for 10-ton scale data capture.
- Led the development of detector control software, integrating sensor arrays and automated emergency alerts.

MITRE Corporation — August 2020 – December 2020
Data Scientist — *Santa Fe, NM*
- Applied complex systems science to analyze satellite network structures and mitigate failure cascades.

– Co-developed **Strategy Mining**, an open-source genetic programming tool for evolving Agent-Based Models (ABMs). *(Source: github.com/mitre/strategy-mining)*
– Conducted semantic analysis of judicial documents to improve legal search accuracy.

Carnegie Mellon University June 2018 – August 2018
Undergraduate Research Fellow *Pittsburgh, PA*
– Executed a research project on the structural analysis of philosophical texts using NLP and Random Walk algorithms; findings published as first author.

Santa Fe Institute June 2017 – June 2018
Undergraduate Researcher *Santa Fe, NM*
– Conducted independent research using NLP and machine learning to explore the narrative function of the chorus in Greek tragedies.

Publications

Preprints

*Equal contribution

– **W.H.W. Thompson**, M. Tian, L. Huynh, A. Wiedemann, V. Modisette, P.S. Chodrow, H.Z. Brooks, "Learning Social Influence with Unobserved Attention." arXiv:2610.12616, 2026. arxiv.org/abs/2610.12616
– A. Chawla*, **W.H.W. Thompson***, J.-G. Young, "Multiple latent orderings better predict language model preferences." arXiv:2609.22170, 2026. arxiv.org/abs/2609.22170
– K. Tang*, D. Kaiser*, **W. Thompson***, J.-G. Young, L. Hébert-Dufresne, N.W. Landry, "Emergent contagion complexity: Disentangling mechanistic complexity from correlated heterogeneity." arXiv:2607.25139, 2026. arxiv.org/abs/2607.25139
– L. Hébert-Dufresne, A. Allard, J.-G. Young, **W.H.W. Thompson**, G. St-Onge, "Simpson's paradox explains the ubiquity of nonlinear, threshold, and complex contagions." arXiv:2605.00791, 2026. arxiv.org/abs/2605.00791
– **W.H.W. Thompson**, Z. Wojtowicz, S. DeDeo, "Levy Flights of the Collective Imagination." arXiv:1812.04013, 2018. arxiv.org/abs/1812.04013

Peer Reviewed Journals

– M.C. Boudreau, **W.H.W. Thompson**, C.M. Danforth, J.-G. Young, L. Hébert-Dufresne, "Sensitivity analysis of epidemic forecasting and spreading on networks with probability generating functions." J. R. Soc. Interface, vol. 23, no. 235, 20250686, 2026. doi:10.1098/rsif.2025.0686
– N.W. Landry, **W.H.W. Thompson**, L. Hébert-Dufresne, J.-G. Young, "Reconstructing networks from simple and complex contagions." Phys. Rev. E, vol. 110, no. 4, L042301, 2024. doi:10.1103/PhysRevE.110.L042301
– K. Adusumilli et al. (incl. **W. Thompson**), "The Structure and Dynamics of Modern United States Federal Case Law." Front. Phys., vol. 9, 695219, 2022. doi:10.3389/fphy.2021.695219

Coherent CAPTAIN-Mills Collaboration

– A.A. Aguilar-Arevalo et al., "Axionlike particle search with a hybrid Cherenkov-scintillation detector." Phys. Rev. D, vol. 114, no. 7, 072008, 2026. doi:10.1103/ztsv-s7ts
– A.A. Aguilar-Arevalo et al., "First Event-by-Event Identification of Cherenkov Radiation from Sub-MeV Particles in Liquid Argon." Phys. Rev. Lett., vol. 135, no. 17, 171804, 2025. doi:10.1103/bklh-w3dn
– A.A. Aguilar-Arevalo et al., "Measurement of the liquid argon scintillation pulse shape using differentiable simulation in the Coherent CAPTAIN-Mills experiment." Phys. Rev. D, vol. 112, no. 7, 072010, 2025. doi:10.1103/34mf-z6l9
– A.A. Aguilar-Arevalo et al., "Testing meson portal dark sector solutions to the MiniBooNE anomaly at the Coherent CAPTAIN-Mills experiment." Phys. Rev. D, vol. 109, no. 9, 095017, 2024. doi:10.1103/PhysRevD.109.095017
– A.A. Aguilar-Arevalo et al., "Prospects for detecting axionlike particles at the Coherent CAPTAIN-Mills experiment." Phys. Rev. D, vol. 107, 095036, 2023. doi:10.1103/PhysRevD.107.095036
– A.A. Aguilar-Arevalo et al., "First Leptophobic Dark Matter Search from Coherent CAPTAIN-Mills." Phys. Rev. Lett., vol. 129, 021801, 2022. doi:10.1103/PhysRevLett.129.021801
– A.A. Aguilar-Arevalo et al., "First dark matter search results from Coherent CAPTAIN-Mills." Phys. Rev. D, vol. 106, 012001, 2022. doi:10.1103/PhysRevD.106.012001

Conference Proceedings

- **W. Thompson** et al., "Evolving Robust Facility Placements." Companion Proc. Genetic and Evolutionary Computation Conference (GECCO), pp. 775–778, ACM, 2023. doi:10.1145/3583133.3590712
- A. Isherwood, **W. Thompson**, et al., "Modeling Macaque Fighting Dynamics with the Evolutionary Model Discovery Framework." Proc. 2022 Conference of the Computational Social Science Society. doi:10.1007/978-3-031-37553-8_9

Grants and Awards

- **Data Science for Social Good Summer Fellowship, Johns Hopkins University**, 2026. $14,000 award; selected from more than 400 applicants. Worked with the City of Baton Rouge on early warning models for blight and structural deterioration.
- **Complexity Award Grant, Vermont Complex Systems Institute, University of Vermont**, 2026. $5,000 for modeling strategic lying in Kenyan pastoralists with Eleonora Zanetti (PhD candidate, Washington State University). Supported survey participant compensation and in-country travel for her June–July 2026 fieldwork in Kenya.
- **Second Place, University of Vermont Computing Student Research Day**, 2023. Presented research on developing learnable asynchronous models of opinion dynamics.
- **Third Place, University of Vermont Computing Science Research Day**, 2022. Awarded for research employing evolutionary algorithms to develop robust facility placements.
- **MITRE Best Paper Incentive Prize**, 2022. For "The Structure and Dynamics of Modern United States Federal Case Law."
- **Large Team Distinguished Performance Award, LANL**, 2022. Recognized for construction and dark matter search leadership with the Coherent CAPTAIN-Mills detector.
- **Science Undergraduate Laboratory Internship (SULI) Grant, DOE**, 2020. Selective DOE grant for post-baccalaureate research at LANL.
- **Dean's Award for College Service, St. John's College**, 2020. Recognized for exceptional leadership and dedication.
- **Robert Neidorf Memorial Scholarship**, 2018.
- **ARIEL Grant, St. John's College**, 2017.

Selected Talks

Invited Seminars

- **W. Thompson**. "The Emergence of Inequality with Two Models of Social Dynamics." Dartmouth College Applied Math Seminar Series, Hanover, NH, Oct. 2023.
- **W. Thompson**. "Understanding Polarization with the Non-Linear Voter Model on Higher Order Networks." Vermont-KIAS Workshop on Higher Order Interactions, Burlington, VT, Sept. 2023.
- **W. Thompson**. "Searching for Light Dark Matter with Coherent CAPTAIN-Mills." University of New Mexico NUPAC Colloquia Series, Albuquerque, NM, May 2021.

Conference Presentations

- **W. Thompson**, A. Chawla, J.-G. Young. "The Parliament of Thoughts in Language Models." IC2S2 2026, Burlington, VT, Jul. 2026.
- **W. Thompson** et al. "Multiple latent preference orderings in language models." Conference on Complex Systems (CCS) 2026 (Oral, contributed session).
- **W. Thompson** et al. "Inferring Interaction Kernels In Stochastic Opinion Dynamics Models." AMS Spring Eastern Sectional Meeting, Special Session on Mathematical Modeling of Ecological and Evolutionary Dynamics, Boston, MA, Mar. 2026.
- **W. Thompson** et al. "Sensitivity of epidemic forecasts with statistical condition estimation." Dynamics Days 2025, Denver, CO (Poster).

- **W. Thompson** et al. "Inferring Interaction Kernels In Stochastic Opinion Dynamics Models." NetSci 2024, Québec City, QC, Jun. 2024.
- **W. Thompson**. "Understanding Polarization In the Higher Order Non-Linear Voter Model." Dynamics Days 2024, Davis, CA, Jan. 2024.
- **W. Thompson**. "The Emergence of Polarization in the Non-Linear Voter Model on Higher Order Networks." Joint Mathematics Meetings, San Francisco, CA, Jan. 2024.
- **W. Thompson** et al. "Evolving Robust Facility Placements." GECCO 2023, Lisbon, Portugal, Jul. 2023.

Teaching & Mentorship

University of Vermont — Burlington, VT
Graduate Research Mentor — *2023 – Present*

- Directed a research cohort of 5 undergraduate interns over two semesters; managed the development of a Machine Learning pipeline to detect student-authored news content across hundreds of U.S. publications.
- Mentored summer interns on individual projects in network science and complex systems.
- **Guest Lecturer:** Delivered lectures on "Introduction to Network Science," "Stochastic Cellular Automata," and "Chaos & Fractals" for graduate and undergraduate courses.

Los Alamos National Laboratory — Los Alamos, NM
Post-Baccalaureate Mentor — *Summers 2021, 2022*

- Supervised undergraduate interns during two consecutive summer sessions, providing technical guidance on C++ development and detector hardware calibration.
- Facilitated onboarding for new researchers, training them on laboratory safety protocols and data acquisition software.

St. John's College — Santa Fe, NM
Head Senior Lab Assistant — *May 2019 – May 2021*

- Directed the setup and instructional design of a tabletop quantum optics lab and trained junior assistants in experimental protocols.

Service & Outreach

- **Peer Reviewer:** PNAS (2026), Physical Review E (2024), NPJ Complexity (2024).
- **Public Outreach:** Delivered talks on "The Mathematics of Making Coffee" and percolation theory at Vermont State University (2024), Vermont Science Olympiad (2024), and UVM MATHCOUNTS (2024).

Technical Skills

Languages: Julia, Python, C++, SQL, Bash, LaTeX
Machine Learning & Data: PyTorch, Flux.jl, NetworkX, SciPy, Pandas, Scikit-learn
Methods: Bayesian Inference, Graph Neural Networks, MCMC, Agent-Based Modeling, NLP
Tools: Git, Docker, UNIX/Linux, High-Performance Computing (HPC)

References

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